Quantum Mechanics II — Exercise Sheet 15

Hydrogen levels and their states

Variant: Questions only

Exercise 1 (Normalizing the second level). Put $a_0 = \hbar^2/(mC)$ and $r = |x|$. The level $-\mathrm{Ry}/4$ contains the real functions

$$f_0 = (1 - r/(2a_0))e^{-r/(2a_0)}, \qquad f_j = x_j e^{-r/(2a_0)}, \quad j = 1, 2, 3.$$

(i) Verify the $l = 0$ and $l = 1$ radial equations for these functions. (ii) Normalize all four with positive real constants and show that they are mutually orthogonal. You may use $\int_0^\infty r^n e^{-r/a_0}\,\mathrm{d}r = n!a_0^{n+1}$ for integers $n \geq 0$.

Exercise 2 (Resolving an angular superposition). For a fixed hydrogen principal number $n \geq 3$ let $|l, \mu\rangle$ denote normalized orbital eigenvectors at that level. Take $\psi = (|0, 0\rangle + \sqrt{2}|1, 0\rangle + \mathrm{i}|2, 1\rangle)/2$. (i) Find the distributions of energy, L^2 and L_3. (ii) Find the vector after retaining $L_3 = 0$, and decide whether that reading fixes l.

Exercise 3 (A line series determines its limit). Neglect recoil and let $\nu_{n\to 2}$ be the photon frequency for hydrogen transitions from principal number $n > 2$ to 2. Find the ratio of the series-limit frequency to $\nu_{3\to 2}$. If the same Coulomb constant is used for a nucleus of mass M and an orbiting particle of mass m_e, state how all these frequencies scale with $m_{\mathrm{red}} = m_e M/(m_e + M)$. Distinguish a limiting frequency from a transition starting in a normalizable state at zero energy.

Exercise 4 (The projected Stark matrix). Use the four normalized real vectors $\phi_0, \phi_1, \phi_2, \phi_3$ of Exercise 1, with their positive constants. Let Π_2 project onto their span and consider the finite matrix of $\Pi_2(Fx_3)\Pi_2$ for $F \in \mathbb{R}$. Compute all its entries and eigenvalues. Interpret the eigenvalues as the formal first-order level shifts only. No isolated eigenvalue or convergent perturbation series is assumed for the full Hamiltonian with the unbounded linear potential.